

LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

PROBLEM STATEMENT 29 : Municipal Infrastructure Damage Reporting App

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WRITTEN JUSTIFICATION:

Architecture Selection – Microservices Architecture

For the Municipal Infrastructure Damage Reporting App, I selected Microservices Architecture. The system is divided into separate services such as Report Management, Authentication, GPS & Photo Processing, Department Routing, Notification Service and Report Repository. These services communicate through defined interfaces.

Reason 1: Different services have different processing requirements. For example, GPS & Photo Processing handles uploaded photographs and extracts location information, while Report Management handles the main report workflow. Keeping them separate allows the photo processing service to be scaled independently when many photographs are uploaded.

Reason 2: The system can continue to provide important functions even if one service has a temporary problem. For example, if the Notification Service is unavailable, reports can still be stored in the Municipal Database and accessed through the Engineer Portal.

Security Advantage

Security is important because the application handles citizen details, photographs, GPS information and municipal reports. The Authentication component is used to verify users and validate tokens. The API Gateway provides a common entry point for requests from the Citizen Application and Engineer Portal. OAuth2/JWT is used for authentication and HTTPS is used to protect data during communication. This helps prevent unauthorized access to reports.

Performance Benefit

The system separates GPS & Photo Processing from Report Management. Therefore, processing photographs does not have to slow down the complete report submission process. Individual services can also be scaled when the number of reports increases. The Offline Storage and Sync Manager also allow citizens to save reports when the network is unavailable and synchronize them after the connection is restored.

Conclusion

The selected architecture fits the application because the main functions can be separated into independent components while still communicating through defined interfaces. The design supports the required features such as report submission, photo upload, GPS extraction, automatic department assignment, engineer updates, offline storage and secure access.

COMPONENT DIAGRAM (USING DRAW.IO)

